

Principles of Microeconomics

Day 2 — Perfect Competition

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Recap and today's plan

Yesterday

We built the cost curves: FC , VC , TC , and the per-unit curves AFC , AVC , ATC , MC .

Today

We add the *revenue* side and let a competitive firm choose how much to produce — when to make output, when to shut down, and where profits end up.

Learning goals

By the end you can

- ① list the features of a perfectly competitive market;
- ② explain why the firm is a *price taker* with $MR = P$;
- ③ find the profit-maximising output from $P = MC$;
- ④ apply the shutdown rule and derive the supply curve;
- ⑤ explain how entry and exit drive long-run profit to zero.

What makes a market “perfectly competitive”?

Four features

- ① Many buyers and many sellers.
- ② A homogeneous (identical) product.
- ③ Free entry and exit in the long run.
- ④ Everyone is well informed about prices.

Real markets are rarely perfect, but crops, currencies and commodities come close.

The firm is a price taker

Consequence of the four features

Each firm is tiny relative to the market, so it *cannot move the price*. It takes the market price P as given.

- Charge a penny more $\Rightarrow$ every buyer goes to a rival (identical product).
- Charge less $\Rightarrow$ pointless, you can already sell all you want at P .
- So the firm faces a *horizontal* demand line at height P .

Revenue for a price taker

Total, average and marginal revenue

$$TR(Q) = P \cdot Q$$

$$AR = \frac{TR}{Q} = P$$

$$MR = \frac{dTR}{dQ} = P$$

Because P is fixed to the firm, every extra unit adds exactly P .

The key revenue fact

For a competitive firm: $MR = AR = P$.

- Selling one more unit brings in P — no more, no less.
- The demand line, the AR line and the MR line are all the *same* horizontal line.
- This is what makes the competitive firm's problem so clean.

The profit-maximising rule

Profit is $\pi(Q) = TR(Q) - TC(Q)$. To maximise, look at the last unit.

Marginal logic

- If $MR > MC$: the next unit adds more revenue than cost — *make it*.
- If $MR < MC$: the next unit costs more than it earns — *do not*.
- Stop where $MR = MC$.

The competitive optimality condition

Since $MR = P$, the rule $MR = MC$ becomes

$$\boxed{P = MC}$$

(on the *rising* part of MC).

- Read it as: produce up to the last unit whose cost equals the price.
- Why the rising branch? On the falling branch, raising output would still add profit — you would not stop there.

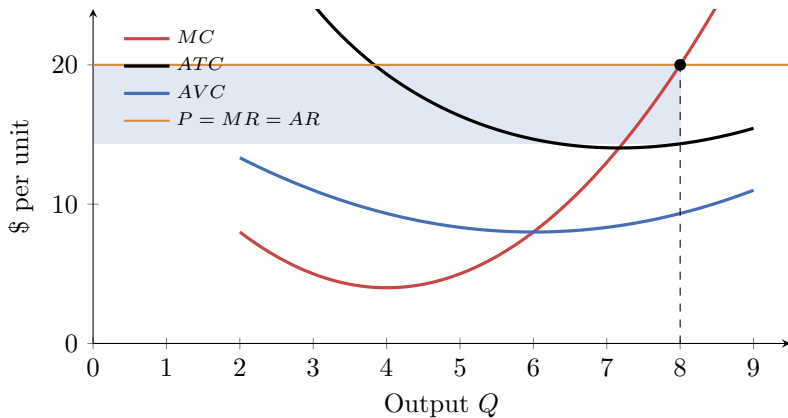
Where does profit show up?

Profit as a rectangle

$$\pi = (P - ATC) \times Q^*.$$

- Height = $P - ATC$ (profit per unit); width = Q^* (units sold).
- If $P > ATC$: positive profit. If $P < ATC$: a loss.
- We can shade this rectangle straight onto the cost diagram.

The competitive firm, drawn

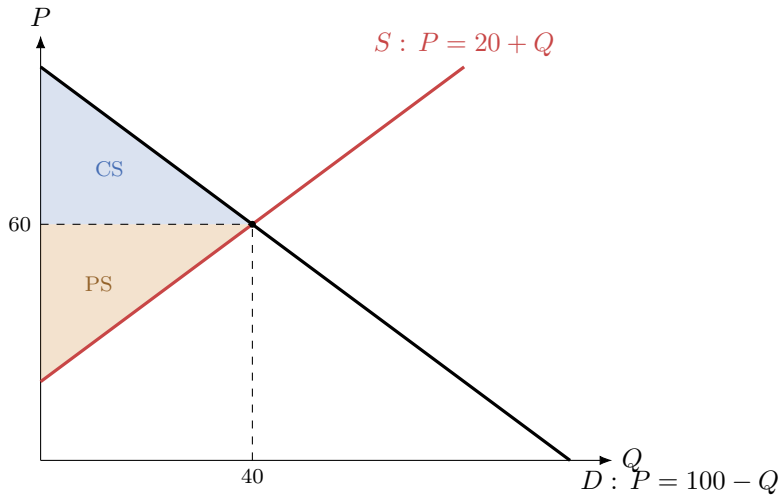


Reading the firm's diagram

Using the same firm as Day 1, with price $P = 20$:

- $P = MC$ on the rising branch at $Q^* = 8$.
- $ATC(8) \approx 14.33$, so profit per unit ≈ 5.67 .
- The shaded rectangle is total profit, $(20 - 14.33) \times 8 \approx 45.3$.
- If the price line dropped below ATC , the rectangle would flip to a loss.

Consumer, producer and total surplus



What the surplus triangles mean

With $D : P = 100 - Q$ and $S : P = 20 + Q$, equilibrium is $(Q, P) = (40, 60)$.

- **Consumer surplus** (blue): what buyers would have paid minus what they did. Here $\frac{1}{2}(100 - 60)(40) = 800$.
- **Producer surplus** (orange): revenue above the cost of supplying. Here $\frac{1}{2}(60 - 20)(40) = 800$.
- Total surplus = 1600, and competition maximises it: no unit is left un-traded that both sides would gain from.

Should the firm operate at all?

Even a loss-making firm might keep going in the short run — because FC is owed anyway.

The comparison

- Operate: lose FC minus any contribution revenue makes toward it.
- Shut down: lose exactly FC (still owed), earn nothing.

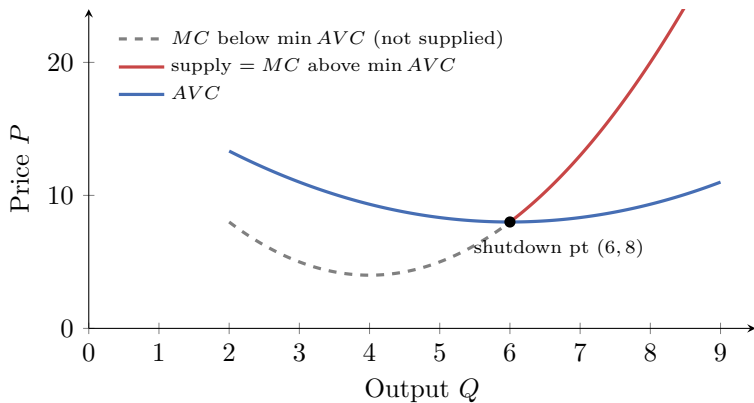
So operate whenever revenue at least covers *variable* cost.

The shutdown rule

Operate in the short run if $P \geq \min AVC$. Otherwise, shut down.

- If $P \geq \min AVC$: revenue covers all variable cost and chips away at FC .
- If $P < \min AVC$: producing loses even more than the fixed cost alone.
- For our Day-1 firm, $\min AVC = 8$ (at $Q = 6$) — so the shutdown price is 8.

The firm's short-run supply curve



Reading the supply curve

- The firm's supply curve is its MC curve — but only the part *above* $\min AVC$.
- Below the shutdown point (grey, dashed) the firm supplies nothing.
- At and above (6, 8) (solid red), quantity supplied rises with price along MC .
- Add up all firms' supply curves horizontally to get *market* supply.

In-slide exercise (no solution given)

Your turn

A competitive firm has $TC(Q) = Q^2 + 4Q + 9$.

- 1 Find MC and AVC .
- 2 What is the shutdown price?
- 3 Write the firm's supply curve $Q(P)$.

Hint: invert $P = MC$ to get Q as a function of P .

Profit is a signal, not a secret

The long-run force

In a competitive market, one firm's profit does not stay private — it advertises an opportunity.

- Positive profit says “there is money to be made here” — outsiders enter.
- A loss says “this is not worth it” — some firms leave.
- Free entry and exit (one of our four features) is what makes this possible.
- The number of firms is therefore *not* fixed in the long run — it adjusts.

Where the adjustment must end

The zero-profit condition

Entry and exit stop only when there is no longer any reason to enter or leave — that is, when profit is exactly zero:

$$P = \min ATC.$$

- Each surviving firm produces at the bottom of its ATC — its efficient scale.
- “Zero profit” means normal returns: the owner still earns what their money and effort could earn elsewhere, just nothing *extra*.
- We now watch this happen, step by step, with numbers.

Our example firm

Every firm in the market is identical, with

$$TC(q) = q^2 + 25 \Rightarrow MC = 2q, \quad AVC = q, \quad ATC = q + \frac{25}{q}.$$

- Efficient scale: $\frac{d ATC}{dq} = 1 - \frac{25}{q^2} = 0 \Rightarrow q = 5$, and $\min ATC = 10$.
- Firm supply: set $P = MC = 2q \Rightarrow q = \frac{P}{2}$.
- Market demand: $Q = 900 - 40P$.

So the long-run price *must* settle at $P = \min ATC = 10$. The question is how it gets there.

A profit formula we can track

Substitute the firm's own supply $q = P/2$ into its profit:

$$\pi = Pq - TC = P \cdot \frac{P}{2} - \left(\left(\frac{P}{2} \right)^2 + 25 \right) = \frac{P^2}{2} - \frac{P^2}{4} - 25 = \frac{P^2}{4} - 25.$$

- Each firm's profit depends only on the *price*.
- $\pi = 0 \iff \frac{P^2}{4} = 25 \iff P = 10$ — exactly min ATC . ✓
- Above $P = 10$: profit; below: loss. Watch P and π move together.

Entry — step 1: too few firms, fat profits

Imagine the industry starts with only **10 firms**.

- Market supply = $10 \times \frac{P}{2} = 5P$. Set equal to demand: $5P = 900 - 40P \Rightarrow P = 20$.
- Each firm produces $q = \frac{20}{2} = 10$; total $Q = 100$.
- Profit per firm = $\frac{20^2}{4} - 25 = 100 - 25 = \boxed{75}$.

Big profits — but with free entry, this cannot last.

Entry — step 2: newcomers pour in

Those \$75 profits attract entrants. As firms join, *market supply shifts right*, and the market price falls. Say the count rises to **40 firms**.

- Supply = $40 \times \frac{P}{2} = 20P$. Then $20P = 900 - 40P \Rightarrow P = 15$.
- Each firm now produces $q = 7.5$; total $Q = 300$.
- Profit per firm = $\frac{15^2}{4} - 25 = 56.25 - 25 = \boxed{31.25}$.

More firms $\Rightarrow$ lower price $\Rightarrow$ thinner profit. Still positive, so entry continues.

Entry — step 3: profit competed away

Entry only stops when profit hits zero — at $P = 10$. How many firms does that take?

- At $P = 10$: each firm produces $q = \frac{10}{2} = 5$ (its efficient scale).
- Total quantity from demand: $Q = 900 - 40(10) = 500$.
- **Number of firms** = $\frac{\text{total output}}{\text{output per firm}} = \frac{500}{5} = \boxed{100}$.
- Profit per firm = $\frac{10^2}{4} - 25 = 0$. No reason for anyone else to enter.

The mirror image: losses and exit

Now suppose instead the industry is *overcrowded* — say **145 firms**.

- Supply = $145 \times \frac{P}{2} = 72.5P$. Then $72.5P = 900 - 40P \Rightarrow P = 8$.
- Each firm produces $q = 4$; profit = $\frac{8^2}{4} - 25 = 16 - 25 = \boxed{-9}$ (a loss).
- Do they shut down now? No: $AVC = q = 4$, and $P = 8 > 4$, so revenue still covers variable cost. They *operate* short-run but bleed money.

A short-run loss is survivable; a permanent one is not. So firms start to leave.

Exit until the loss disappears

As firms exit, *market supply shifts left*, and the price rises.

Firms n	Price P	Output q	Profit per firm
145	8	4.0	-9.00
120	9	4.5	-4.75
100	10	5.0	0.00

Each departure lifts the price a little, which shrinks everyone's loss. Exit stops at $n = 100$, $P = 10$ — the *same* equilibrium the entry story reached from below.

The whole adjustment, at a glance

Firms n	Price P	Profit/firm	Direction	
10	20	+75.00	entry $\downarrow P$	too few firms
40	15	+31.25	entry $\downarrow P$	
100	10	0.00	stop	equilibrium
120	9	-4.75	exit $\uparrow P$	
145	8	-9.00	exit $\uparrow P$	too many firms

Whether we start above or below, the market is pulled to the *same* point: $P = \min ATC = 10$, each firm at $q = 5$, and $n = 100$.

Counting the firms — the recipe

Any long-run competitive market is solved the same way:

Four steps

- ① Long-run price: $P = \min ATC$.
- ② Output per firm: q^* = the output at that minimum.
- ③ Total market output: read Q off demand at that price.
- ④ Number of firms: $n = \frac{Q}{q^*}$.

Entry and exit are the *force*; these four steps are the *arithmetic* of where it lands.

Run or leave? The decision rule

Long-run rule

A firm stays only if the price covers its *full* average cost:

stay if $P \geq \min ATC$, leave if $P < \min ATC$.

- Short-run rule (Day's earlier slide): operate if $P \geq \min AVC$.
- The long-run bar is *higher*: with no fixed cost to protect, covering variable cost is no longer enough — the firm must cover *everything* or exit.

Day 2 — checkpoint

You should now be able to

- explain why a competitive firm has $MR = AR = P$;
- find output from $P = MC$ and draw the profit rectangle;
- state and apply the shutdown rule ($P \geq \min AVC$);
- derive the firm's short-run supply curve;
- explain why long-run profit is competed to zero at $P = \min ATC$.

Tomorrow: remove competition entirely — a single seller with power over price.

In-class exercise

Exercise 2 — Perfect Competition

Work through **Exercise 2** on the sheet: identical firms with a quadratic cost, a demand curve and a fixed number of firms in the short run. You will find the shutdown price, the short-run equilibrium, each firm's profit, and then the long-run price and the number of firms.